

Gregory Spangenberg

Computer Vision Researcher and PhD Candidate

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EXPERIENCE

DATA SCIENTIST | ELEUSIS THERAPEUTICS

Oct 2021 – Jan 2022 | Remote | Contract Part-time

- Built and implemented a data ingestion pipeline to support clinical trial research

RESEARCH ENGINEER | ROTH MCFARLANE HAND AND UPPER LIMB CENTRE

Sept. 2020 – Present | St. Joseph's Health Care London

- Responsible for conducting corporate sponsored orthopedic research
- Led a study to investigate the performance of Wright Medical Group's shoulder implant
- Designed and led a study to validate a surgical navigation system for Intellijoint surgical
- Gathered and analyzed data for LimaCorporate's FDA approval of an orthopedic implant

TEACHING ASSISTANT | DEPARTMENT OF MECHANICAL ENGINEERING

Sept. 2020 – Present | Western University

- Taught "Computational Methods for Mechanical Engineers" an introductory course on programming in Matlab.
- Led tutorial help sessions for students encountering issues with programming

STUDENT RESEARCH ANALYST | ROTH MCFARLANE HAND AND UPPER LIMB CENTRE

May – Aug. 2019 | St. Joseph's Health Care London

- Built a finite element simulation to analyze varus-valgus malpositioning of humeral total shoulder arthroplasty implants.
- Published paper: Humeral Short Stem Varus-Valgus Alignment Affects Bone Stress
- Received the Dean's Award

RESEARCH

PEER-REVIEWED ML PAPERS

- Automatic Determination of the Resection Plane for Shoulder Arthroplasty in Arthritic Humeri : A Deep Learning Model
- Automatic bicipital groove identification in arthritic humeri for preoperative planning: A Random Forest Classifier approach

PHD THESIS

- Preoperative Planning and Complication Prediction for Shoulder Arthroplasty Using Machine Learning

MITACS ACCELERATE FELLOWSHIP

- Draft Paper: Prediction of Acromial Stress Fractures from only Preoperative Imaging
- Draft Paper: Automatic Cortical Thickness Measurement of the Glenohumeral Joint using Deep Learning Segmentation and Landmarking on a Multi-Center External Dataset
- Sponsored by Industry Partner: Exactech Inc.

SKILLS

PROGRAMMING

Proficient:
Python • $\text{\LaTeX}$

Experienced:
R • MATLAB • SQL

Familiar:
C++ • Fortran

LIBRARIES

PyTorch • Monai • Pandas •
Numpy • Plotly • Dash

LIBRARIES MAINTAINED PERSONALLY

shoulder • armcortnet •
armcrop

TOOLS/PLATFORMS

Git • Azure Cloud • Linux

LANGUAGES

English • German

EDUCATION

WESTERN UNIVERSITY

PHD MECHANICAL ENGINEERING
2020 - 2025 | London, Canada
• Collaborative Specialization in Machine Learning in Health and Biomedical Science

WESTERN UNIVERSITY

BESC. MECHANICAL AND MATERIALS ENGINEERING
2016 - 2020 | London, Canada
• 2020 Dean's List
• 2020 Academic All-Canadian Award

SPORT

60M SPRINT HURDLER

WESTERN UNIVERSITY VARSITY TRACK
AND FIELD TEAM
Sept 2017 - May 2023